

Relational Algebra example

Banking Example

branch (branch-name, branch-city, assets)

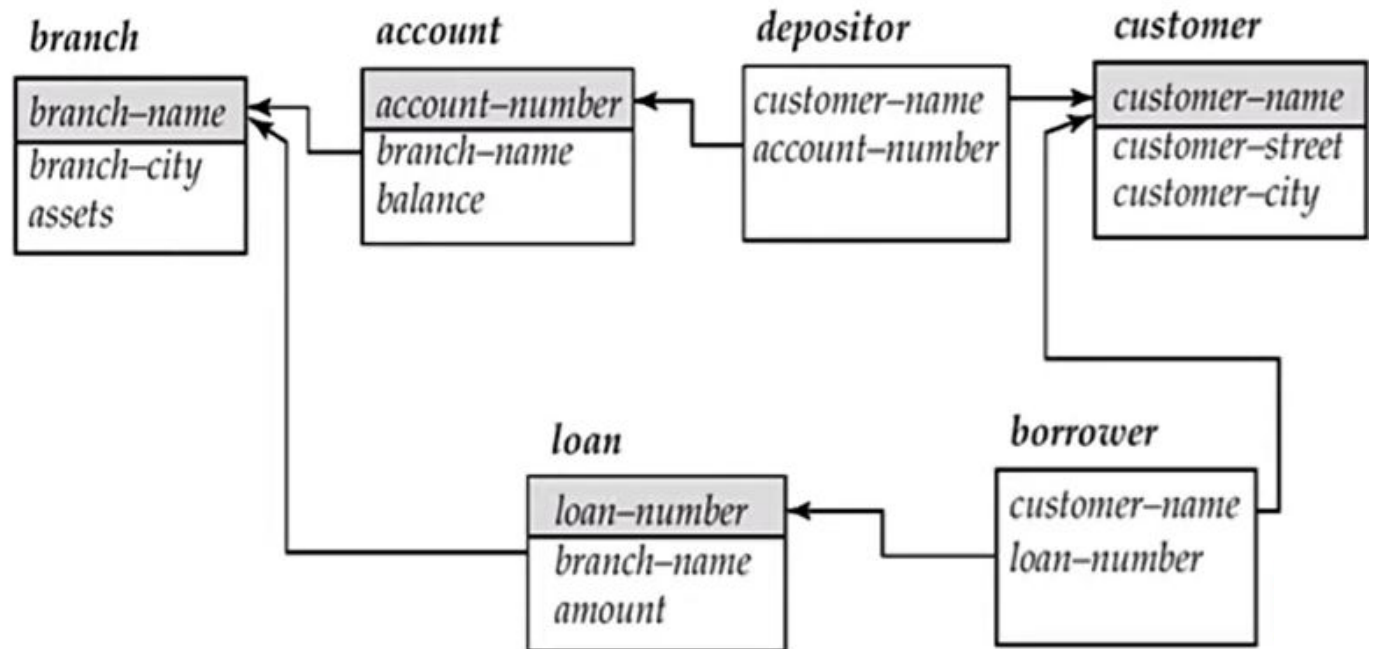
customer (customer-name, customer-street, customer-city)

account (account-number, branch-name, balance)

loan (loan-number, branch-name, amount)

depositor (customer-name, account-number)

borrower (customer-name, loan-number)



branch

<i>branch-name</i>	<i>branch-city</i>	<i>assets</i>
Brighton	Brooklyn	7100000
Downtown	Brooklyn	9000000
Mianus	Horseneck	400000
North Town	Rye	3700000
Perryridge	Horseneck	1700000
Pownal	Bennington	300000
Redwood	Palo Alto	2100000
Round Hill	Horseneck	8000000

customer

<i>customer-name</i>	<i>customer-street</i>	<i>customer-city</i>
Adams	Spring	Pittsfield
Brooks	Senator	Brooklyn
Curry	North	Rye
Glenn	Sand Hill	Woodside
Green	Walnut	Stamford
Hayes	Main	Harrison
Johnson	Alma	Palo Alto
Jones	Main	Harrison
Lindsay	Park	Pittsfield
Smith	North	Rye
Turner	Putnam	Stamford
Williams	Nassau	Princeton

depositor

<i>customer-name</i>	<i>account-number</i>
Hayes	A-102
Johnson	A-101
Johnson	A-201
Jones	A-217
Lindsay	A-222
Smith	A-215
Turner	A-305

account

<i>account-number</i>	<i>branch-name</i>	<i>balance</i>
A-101	Downtown	500
A-102	Perryridge	400
A-201	Brighton	900
A-215	Mianus	700
A-217	Brighton	750
A-222	Redwood	700
A-305	Round Hill	350

loan

<i>loan-number</i>	<i>branch-name</i>	<i>amount</i>
L-11	Round Hill	900
L-14	Downtown	1500
L-15	Perryridge	1500
L-16	Perryridge	1300
L-17	Downtown	1000
L-23	Redwood	2000
L-93	Mianus	500

borrower

<i>customer-name</i>	<i>loan-number</i>
Adams	L-16
Curry	L-93
Hayes	L-15
Jackson	L-14
Jones	L-17
Smith	L-11
Smith	L-23
Williams	L-17

Selection (σ): Example 1

- Find all **loans** made at “Perryridge” **branch**
- Find all **loans** of over \$1200

<i>loan</i>		
<i>loan-number</i>	<i>branch-name</i>	<i>amount</i>
L-11	Round Hill	900
L-14	Downtown	1500
L-15	Perryridge	1500
L-16	Perryridge	1300
L-17	Downtown	1000
L-23	Redwood	2000
L-93	Mianus	500

branch (*branch-name*, *branch-city*, *assets*)

customer (*customer-name*, *customer-street*, *customer-city*)

account (*account-number*, *branch-name*, *balance*)

✓ *loan* (*loan-number*, *branch-name*, *amount*)

depositor (*customer-name*, *account-number*)

borrower (*customer-name*, *loan-number*)

Selection (σ): Example 2

- Find all tuples who have taken **loans** of more than \$1 200 made by the “Perryridge” **branch**

loan

<i>loan-number</i>	<i>branch-name</i>	<i>amount</i>
L-11	Round Hill	900
L-14	Downtown	1500
L-15	Perryridge	1500
L-16	Perryridge	1300
L-17	Downtown	1000
L-23	Redwood	2000
L-93	Mianus	500

Projection (Π): Example 1

- Find all **loan** numbers and the **amount** of the **loans**

loan

<i>loan-number</i>	<i>branch-name</i>	<i>amount</i>
L-11	Round Hill	900
L-14	Downtown	1500
L-15	Perryridge	1500
L-16	Perryridge	1300
L-17	Downtown	1000
L-23	Redwood	2000
L-93	Mianus	500

branch (*branch-name*, *branch-city*, *assets*)

customer (*customer-name*, *customer-street*, *customer-city*)

account (*account-number*, *branch-name*, *balance*)

loan (*loan-number*, *branch-name*, *amount*)

depositor (*customer-name*, *account-number*)

borrower (*customer-name*, *loan-number*)

Projection (Π): Example 2

- Find the **loan** number for each loan of an amount greater than \$1200

loan

<u>loan-number</u>	<u>branch-name</u>	<u>amount</u>
L-11	Round Hill	900
L-14	Downtown	1500
L-15	Perryridge	1500
L-16	Perryridge	1300
L-17	Downtown	1000
L-23	Redwood	2000
L-93	Mianus	500

branch (branch-name, branch-city, assets)

customer (customer-name, customer-street, customer-city)

account (account-number, branch-name, balance)

loan (loan-number, branch-name, amount)

depositor (customer-name, account-number)

borrower (customer-name, loan-number)

Projection (Π): Example 3

- Find those **customers** who lives in “Harrison”

customer

<i>customer-name</i>	<i>customer-street</i>	<i>customer-city</i>
Adams	Spring	Pittsfield
Brooks	Senator	Brooklyn
Curry	North	Rye
Glenn	Sand Hill	Woodside
Green	Walnut	Stamford
Hayes	Main	Harrison
Johnson	Alma	Palo Alto
Jones	Main	Harrison
Lindsay	Park	Pittsfield
Smith	North	Rye
Turner	Putnam	Stamford
Williams	Nassau	Princeton

branch (*branch-name*, *branch-city*, *assets*)

customer (*customer-name*, *customer-street*, *customer-city*)

account (*account-number*, *branch-name*, *balance*)

loan (*loan-number*, *branch-name*, *amount*)

depositor (*customer-name*, *account-number*)

borrower (*customer-name*, *loan-number*)

Union ($\cup$): Example 1

- Find the names of all customers who have a loan, an account, or both, from the bank

```
branch (branch-name, branch-city, assets)
customer(customer-name, customer-street, customer-city)
account(account-number, branch-name, balance)
loan(loan-number, branch-name, amount)
depositor(customer-name, account-number)
borrower(customer-name, loan-number)
```

Intersection($\cap$): Example 1

- Find the names of all customers who have a loan and an account at bank.

```
branch (branch-name, branch-city, assets)
customer(customer-name, customer-street, customer-city)
account(account-number, branch-name, balance)
loan(loan-number, branch-name, amount)
depositor(customer-name, account-number)
borrower(customer-name, loan-number)
```

Set Difference (-): Example 1

- Find the names of all customers who have an account but no loan from the bank.

```
branch (branch-name, branch-city, assets)
customer(customer-name, customer-street, customer-city)
account(account-number, branch-name, balance)
loan(loan-number, branch-name, amount)
depositor(customer-name, account-number)
borrower(customer-name, loan-number)
```

Cartesian Product (x): Example 1

Find the **names** of all customers who have a **loan** at the Perryridge **branch**.

```
branch (branch-name, branch-city, assets)
customer(customer-name, customer-street, customer-city)
account(account-number, branch-name, balance)
loan(loan-number, branch-name, amount)
depositor(customer-name, account-number)
borrower(customer-name, loan-number)
```